Module 02

Preparing Your Laptop

🦞 Ubuntu, Bootable USB, and Everything You Need

OpenClaw Course

🧭 Navigation Chart

By the end of this module, you will be able to:

1. **Verify** your laptop meets the minimum requirements
2. **Explain** why a dedicated Ubuntu laptop is the ideal setup
3. **Create** a bootable Ubuntu USB drive using Rufus
4. **Test** WiFi compatibility before installing
5. **Install** Ubuntu 24.04 LTS (replacing Windows)
6. **Install** Node.js 22+ using nvm
7. **Configure** your laptop for 24/7 operation
8. **Verify** your entire environment is ready for 🦞 OpenClaw

📖 Ship's Logbook *(Part 1)*

Term	Definition
Ubuntu	A beginner-friendly Linux OS — we install version 24.04 LTS on your laptop
ISO	A disk image file containing an entire OS installer — one file, one USB, done
Rufus	A free Windows tool that turns a USB stick into a bootable installer
BIOS/UEFI	Firmware that runs before your OS — access it during startup to change boot order
Boot from USB	Starting your computer from a USB stick instead of the hard drive
LTS	Long-Term Support — 5 years of security updates (24.04 is supported until 2029)

📖 Ship's Logbook *(Part 2)*

Term	Definition
Terminal	Text-based command interface — open with `Ctrl+Alt+T` on Ubuntu
systemd	Linux service manager — works out of the box on Ubuntu (no setup needed)
Daemon	A program that runs continuously in the background
Node.js	JavaScript runtime that OpenClaw is built on — need version 22+
nvm	Node Version Manager — installs and manages Node.js versions
PATH	List of directories your computer checks when you type a command

⚠️ Used or Pre-Owned Laptops

If your laptop came from a **workplace**, check this first:

- **BIOS password** is the only real blocker — if it's locked, you can't boot from USB
- Try: empty password, `admin`, `password`, `1234`, or look up the service tag on **bios-pw.org**
- **MDM / Group Policy / corporate software** — irrelevant, all wiped when Ubuntu is installed

Check	How	Concern
BIOS password	Restart, press BIOS key	**Blocker** if locked
Secure Boot	BIOS settings	Low — can be disabled
Corporate software	N/A	**Wiped** with Windows

🔍 Check Your Hardware: RAM

In Windows (before we erase it), check your RAM:

1. Press `Ctrl + Shift + Esc` to open **Task Manager**
2. Click **Performance** tab > **Memory**
3. Read the number in the top-right corner

RAM	Verdict
4 GB or less	Not enough — OpenClaw will struggle
8 GB	Minimum — works but may be slow
16 GB	Excellent — plenty for everything
32 GB+	Overkill, but great for local models later

🔍 Check Your Hardware: Disk Space

1. Open **File Explorer** (`Windows + E`) > **This PC**
2. Look at your main drive (usually `C:`)

Free Space	Verdict
Less than 10 GB	Not enough
10–20 GB	Tight — will work
20–50 GB	Good
50 GB+	Excellent

After Ubuntu install, space usage:

- Ubuntu + Node.js + OpenClaw: ~15 GB total
- Agent data: grows over time (1–10 GB)

🐙 Why an Old Laptop?

Option	Cost	Privacy	Control
Old laptop	$0–100	Data stays home	You own everything
VPS (cloud)	$5–20/month	Provider's machine	Provider can access
Mac Mini	$500–800	Good, pricey	Overkill for one agent

The old laptop wins on cost, privacy, AND control.

> 🐠 *Raspberry Pi 5 (4 GB+, ~$155 kit) is also viable — runs Linux natively, sips power. But the laptop is simpler for beginners.*

⚠️ Avoid Plug-and-Play + Getting Windows Back

> *"DO NOT use a service that sets up OpenClaw for you. I tried 5 popular services and pretty much all of them exposed the gateway, didn't have pairing mode, and allowed the root directory to be discovered by the internet."*

We set it up ourselves — correctly and securely.

Getting Windows Back (If You Ever Want It)

- Your Windows license is **tied to the motherboard** — it auto-activates on reinstall
- Download the ISO from microsoft.com as insurance before wiping
- Difficulty: **~4/10** — same process as installing Ubuntu, in reverse

📋 What You Will Need

Item	Details
USB stick	12 GB or larger (Ubuntu ISO is ~5.9 GB)
Another computer	To download Ubuntu + create the USB
WiFi password	For the network this laptop will use
Time	60–90 minutes for the full process
Pen and paper	Or phone Notes app — you'll create passwords and tokens to track

You'll also need an **Anthropic API key** for Module 03 — not yet.

> 🔑 *If you haven't created one: go to **console.anthropic.com**, sign up, create an API key.*

⬇️ Step 1: Download Ubuntu and Rufus

On your main computer (not the laptop you're converting):

Ubuntu

1. Go to **ubuntu.com/download/desktop**
2. Download **Ubuntu 24.04.x LTS** (~5.9 GB)

Rufus

1. Go to **rufus.ie**
2. Download **Rufus Portable** (~1.9 MB)

> 💧 *Why 24.04 LTS? Long-Term Support = 5 years of security updates (until 2029). Stable, reliable, perfect for a 24/7 server.*

💾 Step 2: Create Bootable USB with Rufus

1. Plug USB into your **main computer** *(everything on it will be erased!)*
2. Run Rufus (double-click — no install needed)
3. Configure:

Setting	Choose
Device	Your USB stick (check the size!)
Boot selection	Click SELECT → pick the Ubuntu `.iso`
Partition scheme	GPT *(default)*
Everything else	Leave as default

4. Click **START** → accept warnings → wait 5–15 minutes
5. When it says **READY**, click Close — your USB is bootable

⌨ Step 3: Boot from USB

Now move to the old laptop.

1. Plug in the bootable USB and restart the laptop
2. Press the **boot menu key** repeatedly during startup:

Manufacturer	Key
Dell	F12
HP	F9 or Esc
Lenovo	F12
ASUS	Esc or F8

3. Select the **USB drive** from the boot menu

🔭 Step 4: The Ubuntu Installer

After booting from USB, the installer launches. It walks you through setup screens before you commit to anything:

1. **Language** — choose yours, click Continue
2. **Accessibility** — skip unless needed
3. **Keyboard layout** — default is usually correct
4. **Connect to WiFi** — this is your WiFi test! Select your network and connect.

Chipset	Ubuntu Support
Intel	Excellent — almost always works
Qualcomm/Atheros	Good — usually works
Realtek	Hit or miss
Broadcom	Worst — often needs drivers

🔭 Step 4 *(Continued)*: Try or Install

After WiFi, you'll see the choice: **"Try Ubuntu"** or **"Install Ubuntu."**

- **WiFi worked?** → Choose **"Install Ubuntu"** and continue to Step 5
- **Want to explore first?** → Choose **"Try Ubuntu"** to load a full desktop from the USB without changing anything. Install later from the desktop icon.

If WiFi didn't work — three backup plans:

- **USB WiFi adapter** (~$15, Linux-compatible)
- **Phone USB tethering** — plug phone in, enable tethering
- **Ethernet cable** — plug directly into router

🖥️ Step 5: Install Ubuntu *(Key Screens)*

The installer continues with the installation options:

- **Installation type** — choose **Interactive installation**
- **Software selection** — choose **Default selection**

🔑 Check BOTH boxes:

> ***"Install third-party software for graphics and Wi-Fi hardware"*** *— the #1 cause of WiFi not working after installation.*
> ***"Download and install support for additional media formats"*** *— codecs for video and music.*

- **Installation type (disk)** — choose **"Erase disk and install Ubuntu"**

🖥️ Step 5: Install Ubuntu *(Continued)*

Create your account:

- Your name: whatever you like
- Computer name: e.g., `openclaw-laptop`
- Username: **`openclaw`** *(keeps things consistent with the course)*
- Password: choose something memorable — you'll type it often for `sudo`

> 🔑 *Remember this password! There's no "forgot password" link.*

Finish:

- Select your **timezone**
- Click **Install** → wait **15–40 minutes** → click **Restart Now**, remove USB when prompted

📡 Step 6: Switch to Guest WiFi

Your laptop auto-reconnected to the WiFi from the installer. Now let's put it on an **isolated** network.

A **guest network** provides internet but **isolates** this laptop from your other devices. Home WiFi is "flat" — all devices see each other. A compromise could spread.

Set up a guest network *(from your phone or main computer):*

1. Open your router's admin page (Default Gateway address in browser)
2. Find **Guest Network** settings → enable it, set a password
3. Ensure **"Allow guests to access local network"** is **OFF**

Connect the laptop: Top-right system menu → **Wi-Fi** → connect to the **guest** network

📦 Step 7: Update Ubuntu and Install Tools

Open a terminal: click **Terminal** in the dock or press `Ctrl+Alt+T`

```
sudo apt update && sudo apt upgrade -y
```

First run takes 5–10 minutes. Password won't show when you type — that's normal.

Then install essential tools:

```
sudo apt install -y git curl wget build-essential
```

Tool	Purpose
`git`	Version control and backups
`curl` / `wget`	Download files (used by installers)
`build-essential`	Compilation tools for npm packages

Step 8: Install Node.js 22

```
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.40.4/install.sh | bash
```

Close the terminal, reopen it (`Ctrl+Alt+T`), then:

```
nvm install 22
nvm alias default 22
```

Verify:

```
node --version    # Should show: v22.x.x
npm --version     # Should show: 10.x.x or 11.x.x
```

🔋 Step 9: Power Settings for 24/7 Operation

GUI: Disable auto-suspend

1. Open **Settings** > **Power**
2. Set **Automatic Suspend** → **OFF** (both battery and plugged in)
3. Set **Screen Blank** → **Never**

Terminal: Lid close behavior

```
sudo sed -i 's/#HandleLidSwitch=suspend/HandleLidSwitch=ignore/' /etc/systemd/logind.conf
sudo sed -i 's/#HandleLidSwitchExternalPower=suspend/HandleLidSwitchExternalPower=ignore/' /etc/systemd/logind.conf
sudo systemctl restart systemd-logind
```

> ⚠️ *The restart command logs you out. Screen may go black for 1–2 minutes on older laptops — this is normal. If stuck, hold the power button 5 seconds to restart. Test: close lid, wait 30 seconds, open — if it didn't sleep, you're good.*

Shoals and Sandbars

Problem	Fix
Laptop won't boot from USB	Check BIOS key (table in Step 3). Try disabling Secure Boot
WiFi not working	Did you check "third-party drivers" during install? Try `sudo ubuntu-drivers autoinstall`. Use USB adapter, phone tethering, or Ethernet as backup
BIOS password locked	Try defaults, bios-pw.org, or use a different laptop
`node: command not found`	Close terminal and reopen. Or: `source ~/.bashrc`
Laptop sleeps despite settings	Check both Settings > Power AND logind.conf (Step 9)
Password not accepted for `sudo`	Nothing shows when you type — that's normal. Type carefully

✅ Verification Checklist

Run these in the terminal to confirm everything is ready:

```
systemctl is-system-running     # Should say: running
node --version                  # Should say: v22.x.x
npm --version                   # Should say: 10.x.x or 11.x.x
git --version                   # Should say: git version 2.x.x
curl --version | head -1        # Should show curl version
pwd                             # Should say: /home/openclaw
```

If all six pass — your environment is ready for Module 03.

> 🐙 *On native Ubuntu, systemd works out of the box. No config file to write, no restart dance. It's already running.*

🦀 Hands on Deck

Complete this checklist before moving to Module 03:

- [ ] RAM confirmed ≥ 8 GB
- [ ] Ubuntu 24.04 LTS installed and booting
- [ ] WiFi connected (guest network if possible)
- [ ] `systemctl is-system-running` → running
- [ ] `node --version` → v22.x.x
- [ ] `npm --version` → 10.x.x+
- [ ] `git --version` → version number
- [ ] Power: auto-suspend OFF, lid close = ignore
- [ ] `pwd` → `/home/openclaw`

🐙 Treasure Chest

1. **An old laptop is the best OpenClaw host** — $0 cost, full privacy, full control
2. **Ubuntu replaces Windows** — native Linux is simpler than running Linux inside Windows
3. **You can always get Windows back** — license is tied to the motherboard
4. **The installer tests WiFi before you commit** — connect during setup, before choosing Install
5. **Third-party drivers checkbox is critical** — determines if WiFi works after install
6. **systemd works out of the box** — no configuration needed on native Ubuntu
7. **Node.js 22+ via nvm** — easy version management
8. **Disable auto-suspend AND lid close** — both settings, or the laptop sleeps

⛵ Next Port of Call

Module 03: Installing OpenClaw

> *Your laptop is ready. Ubuntu is running. Node.js is installed. Time to bring your agent to life.*